

- 4 (a) Explain what is meant by an *ideal* gas.

An ideal gas is one that obeys the ideal equation $PV = nRT$ for all values of pressure, P , volume, V and thermodynamic temperature, T for a fixed mass of gas where R is the molar gas constant, n is number of moles of gas which is a constant for a fixed mass of gas.

- (b) A cylinder contains 3.16 mol of an ideal gas at a pressure of 4.81×10^5 Pa and a volume of 1.20×10^{-2} m³. Heat is removed for a duration of 12 minutes at constant pressure such that its temperature decreased by 110 K.

- (i) Calculate the initial temperature of the gas.

$$PV=nRT, T_1 = P_1V_1/nR = (4.81 \times 10^5)(1.20 \times 10^{-2})/(3.16)(8.31) = 220\text{K} \\ (219.8 \text{ K})$$

initial temperature = K [1]

- (ii) Determine the new volume of the gas after heat is removed.

For a fixed mass of gas at constant pressure, $\frac{V_2}{V_1} = \frac{T_2}{T_1}$

$$V_2 = \frac{T_2}{T_1} V_1 = \frac{110}{220} (1.20 \times 10^{-2}) = 6.00 \times 10^{-3} \text{ m}^3$$

volume of the gas = m³ [2]

- (iii) Determine the change in internal energy of the gas.

Loss in internal energy, $\Delta U = 3/2 nR\Delta T$ (or $3/2 P\Delta V$, ΔV is –ve due to a decrease in volume.)

$$\Delta U = \frac{3}{2} (3.16)(8.31)(-110) = -4330 \text{ J}$$

change in internal energy = J [1]

(iv) Determine the rate of heat loss.

From first law of thermodynamics, $\Delta U = Q + WD$

$$WD = P\Delta V = (4.81 \times 10^5)(1.20 \times 10^{-2} - 6.00 \times 10^{-3}) = 2890 \text{ J (2886J)}$$

$$Q = -4330 - 2890 = -7220 \text{ J (-7222 J)}$$

Since the Heat supplied is -7220J, Heat lost = + 7220 J

$$\text{Rate of heat loss} = 7220 / (12 \times 60) = 10 \text{ W (10.0 W)}$$

rate of heat loss = W [3]

[Total: 9]